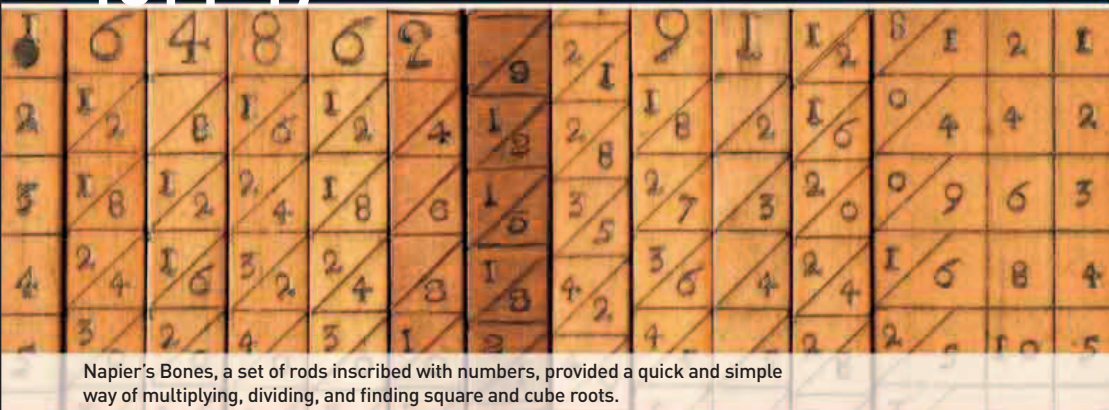




Napier's Bones, a set of rods inscribed with numbers, provided a quick and simple way of multiplying, dividing, and finding square and cube roots.

IN 1614, SCOTTISH MATHEMATICIAN JOHN NAPIER

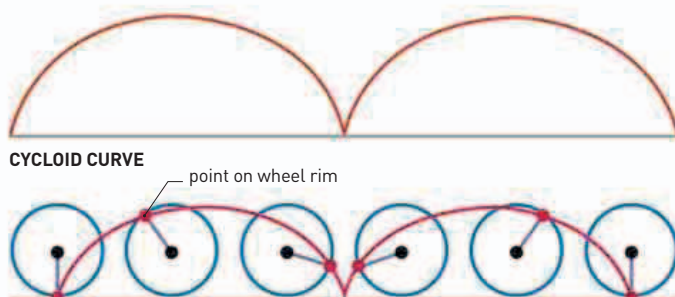
(1550–1617) published a description of logarithms (a logarithm is the power to which a base, such as 10, must be raised to produce a given number), showing how they could resolve “the tedium of lengthy multiplications and divisions, the finding of ratios, and... the extraction of square and cube roots.” The logarithm of the product of two numbers multiplied together is equal to the sum of each number's logarithms. Using tables that Napier included in his book, it was possible to find the product of two numbers by looking up their logarithms, adding them, and then looking up the result in a table of antilogarithms.

Santorio Santorio (1561–1636), also known as Sanctorius, Professor of Anatomy in Padua, Italy, described his experiments in the **study of metabolism** in *De Statica Medicina* (*On Medical*

3,959

MILES

THE MEAN RADIUS OF EARTH



CYCLOID CURVE

point on wheel rim

WHEEL ROLLING IN STRAIGHT LINE

Measurement). Over 30 years, he maintained a record of his weight, and the weights of everything he ate and drank, and of all feces and urine that he passed, and discovered a discrepancy that he attributed to “insensible perspiration.”

In 1615, French mathematician and theologian, **Marin Mersenne** (1588–1648), was the first to properly define the cycloid curve traced by a point on the rim of a wheel. He also made an unsuccessful attempt to calculate the area under the curve, and so posed a problem that several 17th-century mathematicians tried to solve.

In April 1616, for the first of his annual lectures at the Royal College of Physicians in London, England, **William Harvey** (1578–1657) spoke on the **circulation of blood**. He was the first to explain the way the heart pumps oxygenated blood around the body. During this seven-year

Cycloid

The curve described by a point on the rim of a circular wheel as it rolls on a flat surface, known as a cycloid, fascinated mathematicians in the 17th century.

lecture series he expounded on his theory, but he did not publish a full account until 1628.

In 1617, Dutch astronomer and mathematician **Willebrord Snellius** (1580–1626) published his work *Eratosthenes Batavus* (*The Dutch Eratosthenes*), in which he described a new method for **measuring Earth's radius**, by first finding, using triangulation, the distance between two points separated by just one degree of latitude. His work is seen as the foundation for modern geodesy—surveying and measuring Earth.

This year, Napier presented another aid to calculation in his book *Rabdologiae* (meaning measuring rods). This was a set of rods inscribed with figures derived from multiplication tables, which became known as **Napier's Bones**.



This reconstruction of Drebbel's submarine, the first navigable underwater craft, is complete with fins, a rudder, and watertight portholes for oars.

“MY METHOD, THOUGH HARD TO PRACTICE, IS **EASY TO EXPLAIN**... I PROPOSE TO ESTABLISH PROGRESSIVE STAGES OF **CERTAINTY**.”

Francis Bacon, English philosopher, from *Novum Organum*, 1620

IN 1619, German astronomer **Johannes Kepler** (see 1601–04) published *Harmonices Mundi* (*The Harmony of the World*). In this book, he explained the structure and proportions of the universe in terms of geometric shapes and musical harmonies, in much the same way as the ancient philosophers Pythagoras and Ptolemy had done before him. Much of Kepler's thesis concerned the **harmony of the spheres**—the idea that each planet produces a unique sound based on its orbit. It also discussed the relationships of astrological aspects (the angles between planets) and musical tones. More influentially, the thesis contained in its final section a statement of **Kepler's third law of planetary motion**, describing the relationship between a planet's distance from the Sun and the time taken to orbit around it, and the speed of the planet at any time in that orbit (see pp.100–01).

Dutch inventor **Cornelis Drebbel** (1572–1633) had moved to England around 1604. In 1620,

he invented the **first navigable submarine**, while in the employment of the British Royal Navy. It was based on a design suggested by British writer



FRANCIS BACON
(1561–1626)

Born into an aristocratic family, Francis Bacon studied at Trinity College, Cambridge, UK, from age 12. A lawyer and Member of Parliament, he was knighted by James I, who appointed him Attorney General (1613) and Lord Chancellor (1618). In 1621, Bacon was found guilty of corruption. He devoted the rest of his life to writing.

1614 Scottish mathematician John Napier publishes *The Description of the Wonderful Canon of Logarithms*

1615 French mathematician Marin Mersenne describes the cycloid curve

1617 Dutch mathematician Willebrord Snellius publishes the first modern work on the surveying and measurement of Earth

1619 German astronomer Johannes Kepler proposes his third law of planetary motion in *The Harmony of the World*

1614 Italian physiologist Santorio Santorio publishes *On Medical Measurement*, with descriptions of metabolism

1616 English physician William Harvey discusses the circulation of blood in his lectures at the Royal College of Physicians

1617 John Napier describes his calculating device—**Napier's Bones**—in *Rabdologiae*